Preliminary Investigation into the Proton Spin Sensitivity in a Quasi-Frozen Spin Lattice for Axion Research

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Abstract. The work presents preliminary results of a theoretical analysis and modeling of proton spin sensitivity in a quasi-frozen spin lattice, used as a broadband axion antenna, in view of the axion field local phase coherence time and the polarized beam spin coherence time restrictions. Three lines of investigation include: axion field action resonance conditions; the proposal to use (quasi-)frozen spin technologies to improve the axion field local phase coherence time; the conditions imposed by sextupole fields used to improve the beam spin coherence time.

1. Introduction

The detection of axions and axion-like particles in laboratory conditions runs into the problem of their extremely weak interaction with one another and other subatomic particles. However, particle spins prove sensitive to axion presence.

Two axion-spin coupling mechanisms of interest in the context of particle accelerators with polarized beams are, in the first place, the particle's development of an oscillating electric dipole moment (EDM) and, in the second, the spin's sensitivity to the axion field gradient (which in the accelerator is always directed tangentially to the closed orbit and acts like a distributed RF-solenoid at the axion field frequency): the so-called "pseudo-magnetic" effect.

In both cases, axion field detection is based on observing the resonance which appears when the g-2 and axion field frequencies coincide. Therefore the method of investigation into the domain of axion and axion-like particles in a storage ring is led toward scanning a g-2 frequency range, with the aim of crossing the point of consonance between the g-2 precession and axion oscillation. [1] Consider a pure-magnetic ring (Fig. 1). The spin vector precesses about the direction set by the guiding magnetic field (vertical) with a frequency $\omega_{spin} = \gamma G \omega_{cyc}$ due to its magnetic dipole moment (MDM). This we designate the g-2 precession. If an EDM is also present, it will be driving the spin-vector into a precession about the radial axis ($\hat{x}$). However, if the EDM is constant (case a), once the spin-vector orientation changes 180° as a result of its g-2 precession, the EDM-action will turn from driving the spin-vector into appearance to driving its disappearance. On the other hand (case b), if the EDM oscillates at the same frequency as the g-2 precession the two processes become synergetic and constitute a resonance condition. The pseudomagnetic case is analogous, with the substitution of the EDM by a solenoidal field.

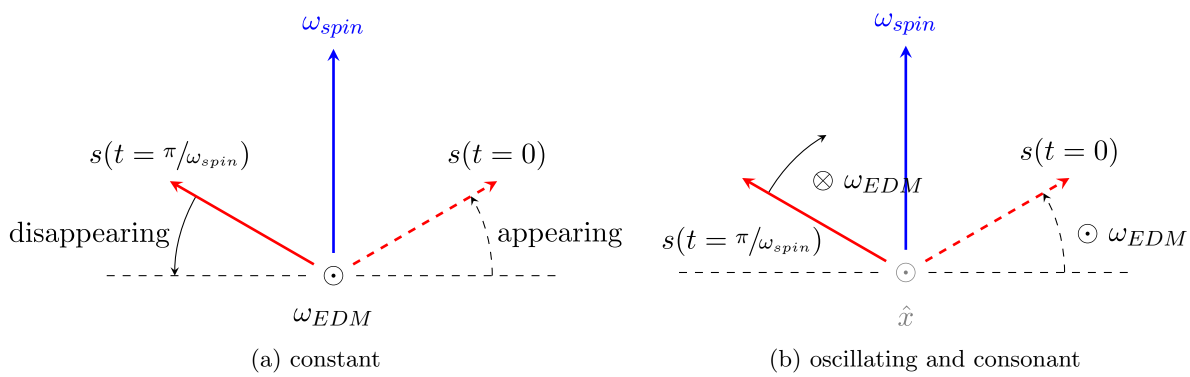


Figure 1: EDM action on a freely-precessing spin vector in a pure-magnetic ring.

The following investigation employs two computational models. In section 2.1 the model provided in [2] has been used. For the investigation in section 2.2 an original model has been developed, based on the spin-rotation representation of spin dynamics in storage rings. [3]

2. Investigation

2.1. Resonance power

By resonance power we understand the maximal amplitude of spin oscillations driven by the axion field (by either of the two mechanisms). Consequently, the g-2 and axion field consonance has the resonance power 1. Any deviation $\Delta\omega = \omega_{axion} - \omega_{MDM}$ from consonance diminishes the potential for the vertical polarization projection growth (see Fig. 2).

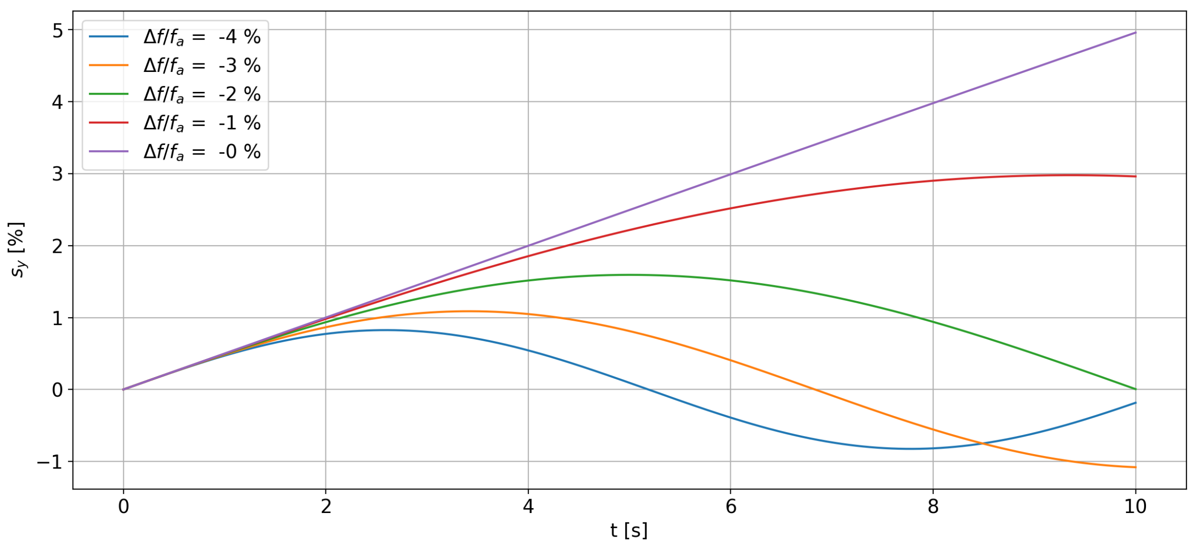


Figure 2: The vertical polarization projection growth in the vicinity of consonance. (Used model parameters are arbitrary and do not reflect expected effect magnitudes.)

Accordingly, the resonance *power* (1) has a bell shape and (2) is dependent on the axion field phase at which the beam was injected. The resonance power *expression* also depends on (3) the experiment duration T, limited by the axion field phase coherence time. The bell shape is characterized by a width $\Delta f = \frac{0.8}{T}$, determined through an 80% decrease in resonance potential. See Figs. 3 and 4 for the relevant simulation results.

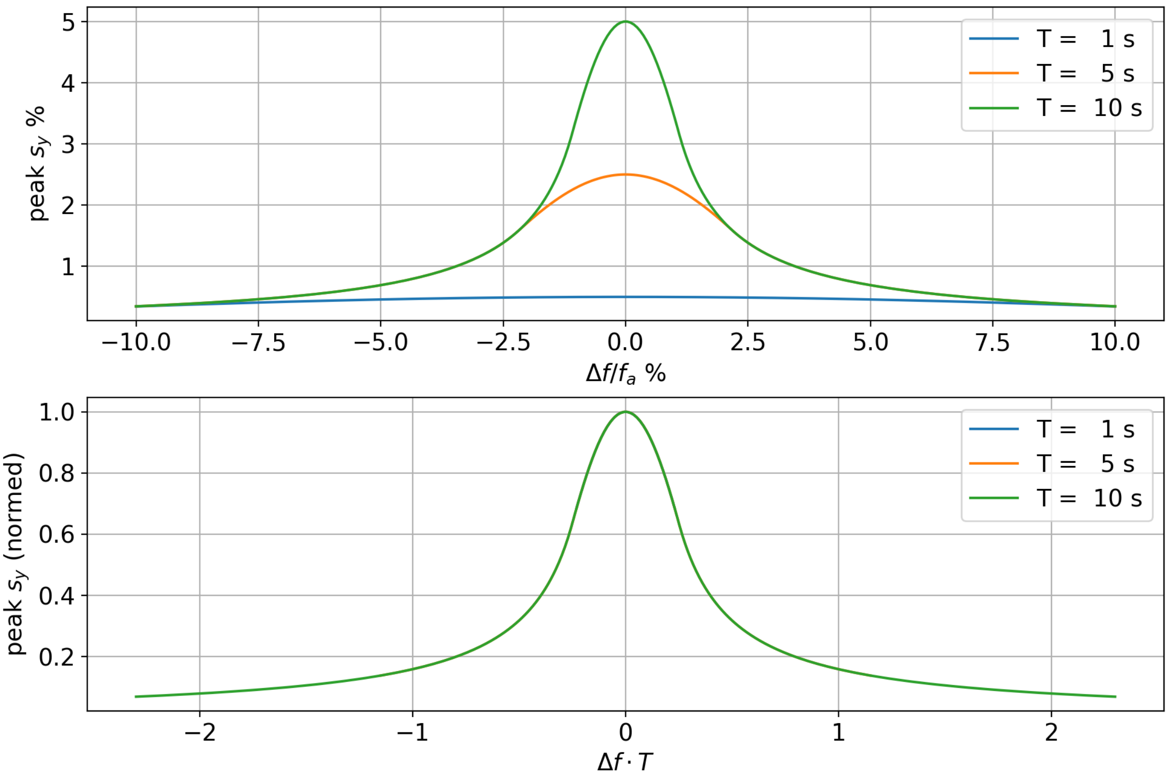


Figure 3: The resonance power dependence on the g-2 frequency. **Top panel**: the s_y accumulated during experiment time T as a function of the deviation $\Delta\omega = \omega_{axion} - \omega_{spin}$. **Bottom panel:** same but normalized.

2.2. Low-frequency measurements

As has been stated in the previous section, the expression of resonance power is limited by the axion field phase coherence time $\tau_a = \frac{h}{m_a v^2}$. Here v is the maximal velocity of the galaxy-bound axions, $m_a = \frac{\hbar\omega_a}{c^2}$ their mass, ω_a the field frequency to be detected. One can see that $\tau_a \propto 1/\omega_a$. This gives an idea to shift to a low frequency spectrum to detect axions forming a more stable field.

To do that, the concept of a quasi-frozen spin ring [4] has been proposed, a modification of the frozen spin ring concept [5, 6] allowing for implementation in an existing storage ring.

In the more general terms this concept can be represented as the neutralization of spin-driving forces in an even number of tacts. Conceived in this way, the 8-shaped ring [7] is also a species. Since the neutralization in this ring is achieved through topology alone, i.e. optical elements of the same kind (magnetic dipoles), having an identical spin-tune energy dependence $\nu(\gamma)$, are used, it has a potential for outstandingly long spin coherence times. To put it in another way, the topological neutralization of the g-2 precession removes the incoherent part of spin resonance responsible for spin decoherence: betatron-oscillating particles, though having an elevated equilibrium energy γ, remain neutralized. Alternatively, one can use a lattice scheme in which the arc dipole action is compensated by Wien filters in the straight section

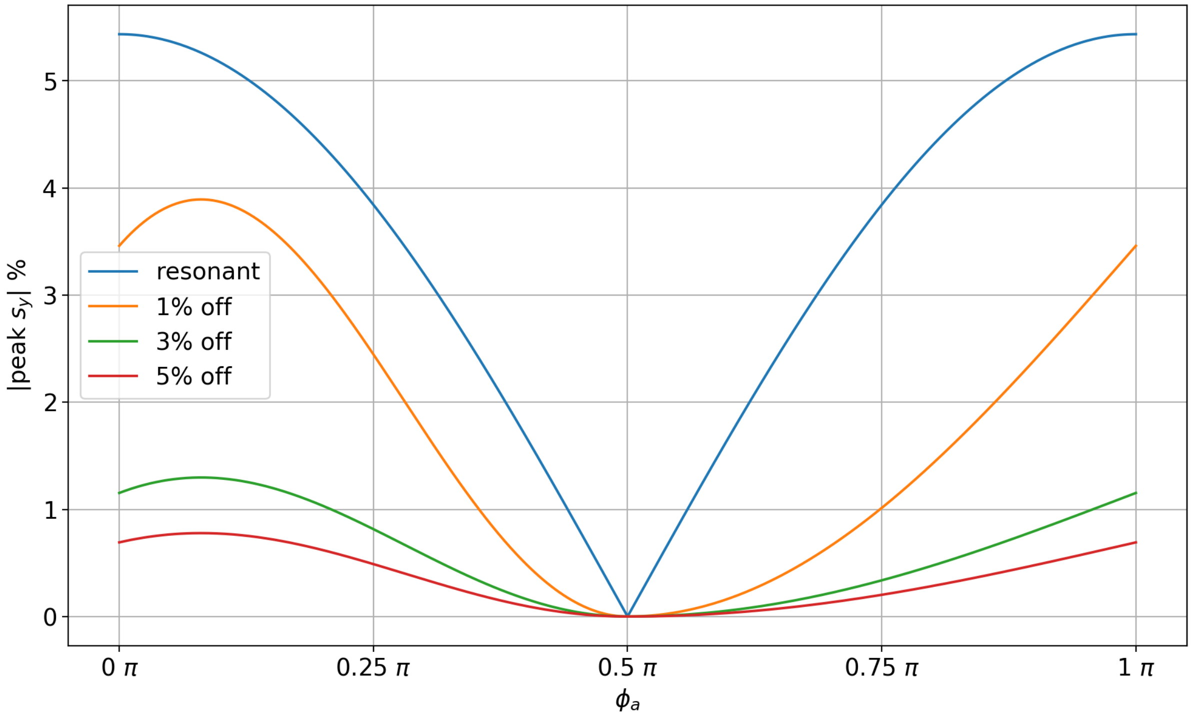


Figure 4: The resonance power dependence on the axion field phase. (The resonance-representing curve exceeds 5%, inconsistent with the other plots, on account of the experiment duration increase, from T = 10 sec to 10.87 sec, introduced in order to let the 1%-off curve reach the peak magnitude.)

(or electrostatic deflectors); but the spin-dynamical element heterogeneity introduced calls for additional measures to suppress spin decoherence.

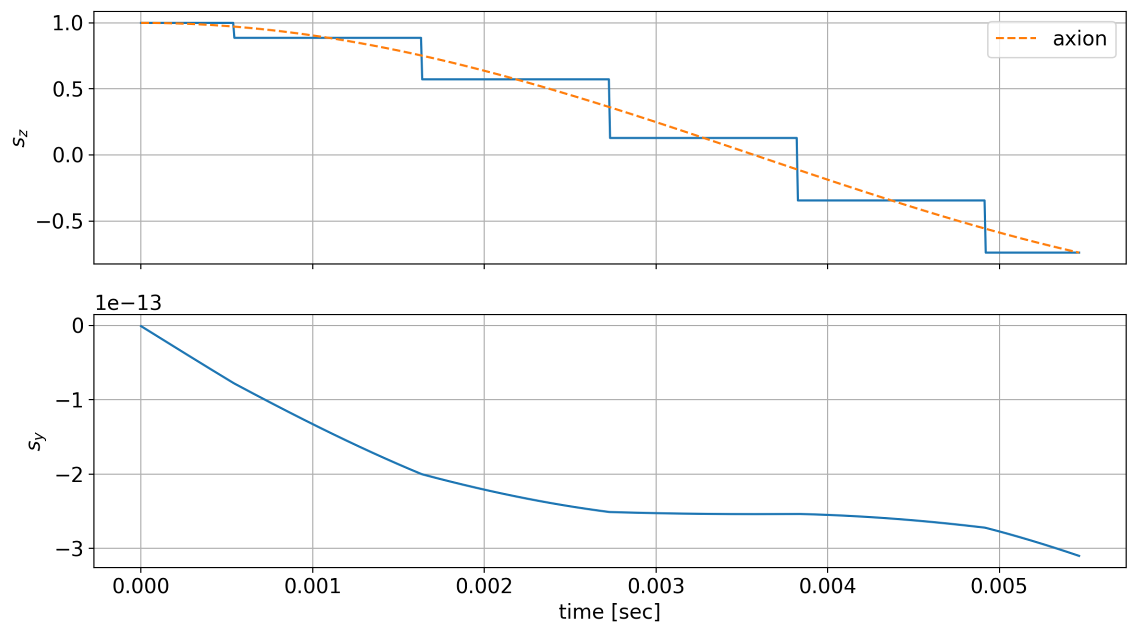
Working in a quasi-frozen setup has certain advantages in respect of the axion project. One is of course the possibility to lower the scanning speed, allowing for the accumulation of a more sizeable signal (the expression of resonance power).

The other one is that the beam energy can remain fixed. The g-2 frequency variation is done by a separate Wien filter scanner (in both the 8-ring and the racetrack-QFS one). Figures 5 and 6 show the results of spin-tracking, when the WF is attuned to the axion field frequency. The first of these compares the strictly friozen with a quasi-frozen cases, the latter exhibits the resonance to be observed.

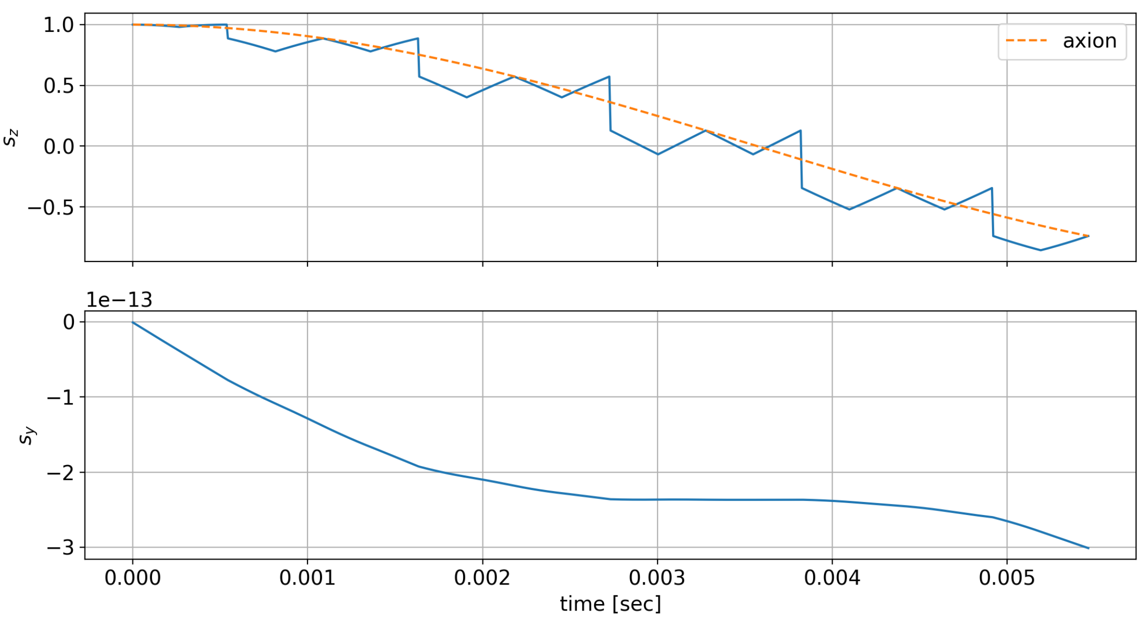
2.3. The role of sextupole fields

When using the racetrack-QFS scheme (the spin-dynamically heterogenous one), one has to utilize sextupole fields to suppress spin-decoherence (thus, to prolong the polarization lifetime). The theory behind their action has been laid out in [8], the effectiveness confirmed experimentally in [9]. (A further work on the influence of spin resonances on the spin coherence time is presented in [10].) Briefly stated, sextupole fields effect an equality of the equilibrium level energies γ_{eq} of beam particles. Since the spin precession frequency $\omega_{spin} = \gamma_{eq} G \omega_{cyc}$, the beam particles' spin-vectors precess at a single frequency (coherently) as a result. This unifying effect is conducive to the accumulation of the axion phase.

Sextupoles aren't required in the case of the figure 8 ring.



(a) The strict FS lattice. The steps in the upper panel reflect the action of the WF scanner. After that the spin keeps its orientation relative to the momentum.



(b) The quasi-FS lattice. Unlike the previous case the spin vector wobbles in-between the scanner action.

Figure 5: Spin-tracking in the low-frequency/low-mass domain. The beam has made five turns inside the ring. The dashed line represents the axion field oscillation. The model parameters are arbitrary: the cyclotron and axion field frequencies respectively 915 and 70 Hz, the maximal spin precession frequency due to the axion field $\omega_x = 1.43 \cdot 10^{-10}$ rad/sec.

3. ACKNOWLEDGMENTS

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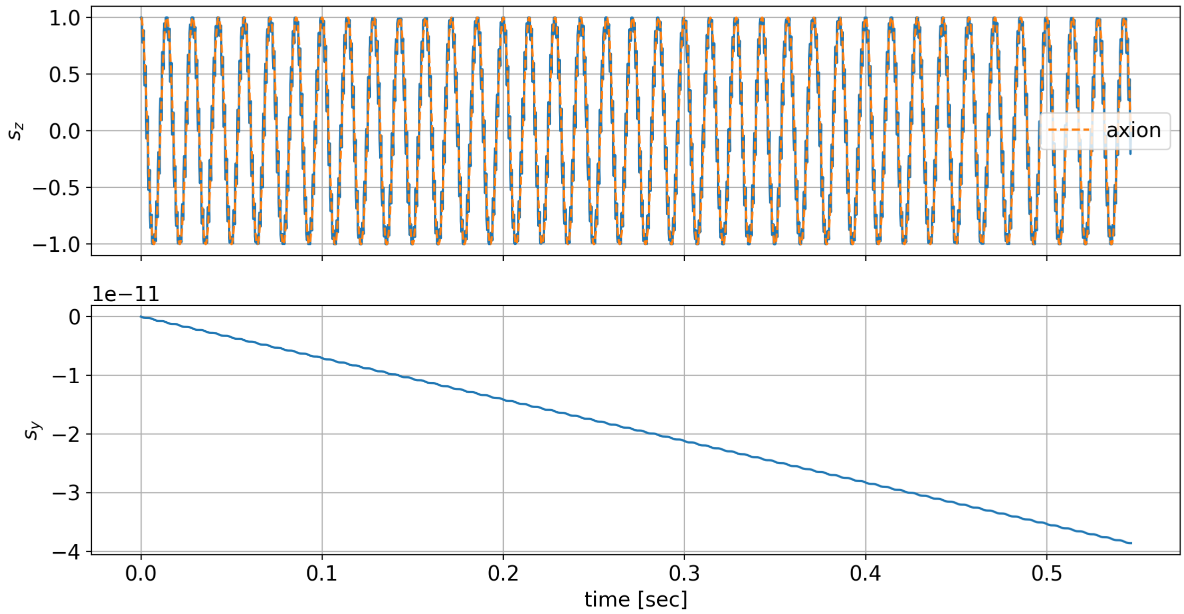

Figure 6: Spin-tracking in the quasi-frozen spin lattice for 500 turns. (Same model parameters as in the previous figure.)

REFERENCES

[1] S. Karanth et al. (JEDI Collaboration), "First Search for Axion-Like Particles in a Storage Ring Using a Polarized Deuteron Beam," Phys. Rev. X, vol. 13, p. 031004, 2023.

[2] J. Pretz et al., "Statistical sensitivity estimates for oscillating electric dipole moment measurements in storage rings," Eur. Phys. J. C, vol. 80, p. 107, 2020.

[3] A. Melnikov, A. Aksentyev, Yu. Senichev, S. Kolokolchikov, "Quasi-Frozen Spin Concept at NICA for EDM Search and Its Matrix Analysis," Phys. Atom. Nucl., vol. 87(4), p. 447-450, 2024.

[4] Y. Senichev et al., Quasi-frozen spin concept of magneto-optical structure of NICA adapted to study the electric dipole moment of the deuteron and to search for the axion, in Proceedings of the 13th International Particle Accelerator Conference (IPAC2022), p. 492.

[5] `https://www.bnl.gov/edm/files/pdf/deuteron_proposal_080423_final.pdf`, retrieved 2nd July 2025.

[6] `https://www.bnl.gov/edm/files/pdf/proton_edm_proposal_20111027_final.pdf`, retrieved 2nd July 2025.

[7] R. Suleiman, Y. Derbenev, M. Grau, V. Morozov, Particle accelerator spin-transparent storage rings for beyond state-of-the-art science, in *Proceedings of the 15th International Particle Accelerator Conference* (IPAC2024), p. 1897.

[8] Yu. Senichev et al. (JEDI Collaboration), Spin tune decoherence in multipole fields, in Proceedings of the 5th International Particle Accelerator Conference (IPAC2014), p. 3017.

[9] G. Guidoboni et al. (JEDI Collaboration), "How to Reach a Thousand Second in-Plane Polarization Lifetime with 0.97-GeV/c Deuterons in a Storage Ring," Phys. Rev. Lett., vol. 117, p. 05548801, 2016.

[10] A. Melnikov et al., "Studies of the spin coherence time of protons at COSY," J. Phys.: Conf. Ser., vol. 2687, p. 052025, 2024.